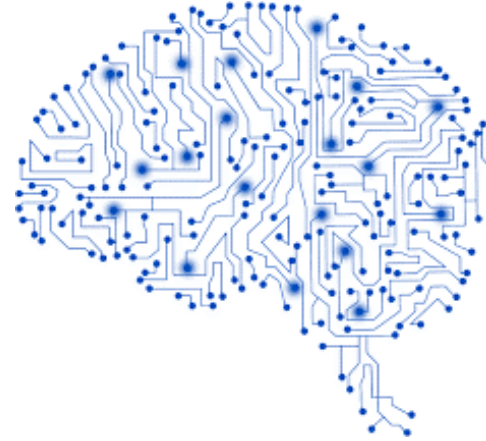




University of Minho
School of Engineering



Dados e Aprendizagem Automática

Linear and Logistic Regression

DAA 2025/2026

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Part IV

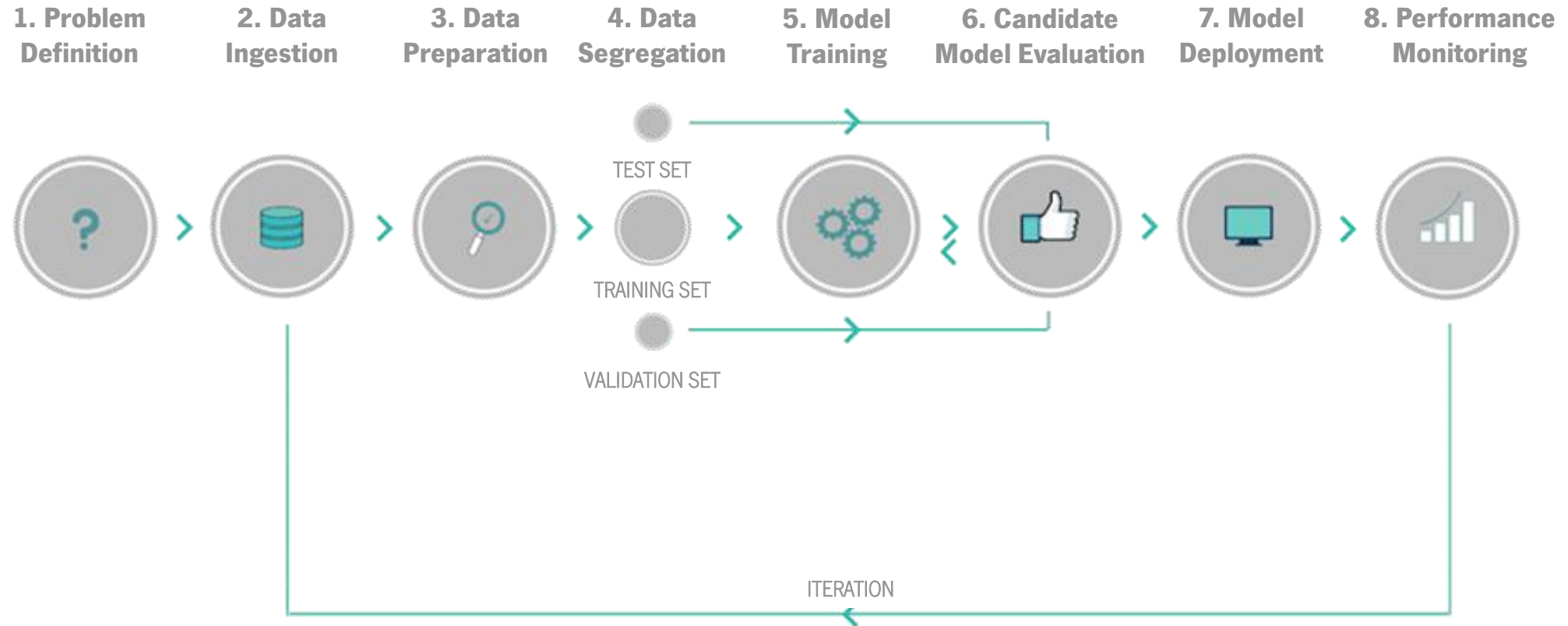
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- Linear Regression
- Logistic Regression
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Supervised Learning

3



We will explore the **Supervised Learning** algorithms.

The choice of the algorithm to implement depends on the type of data and the context provided. If we have labelled data or know the desired output in advance, we use the supervised learning paradigm.



Linear Regression

The Problem and the Data

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Problem: development of a Machine Learning model capable of **predicting house prices** in regions across the USA.

Approach: **Linear Regression** approach.

Dataset: information about houses in regions of the United States, including:

- ***Avg. Area Income***: average income of residents in the city where the house is located;
- ***Avg. Area House Age***: average age of houses in the same city;
- ***Avg. Area Number of Rooms***: average number of rooms in houses in the same city;
- ***Avg. Area Number of Bedrooms***: average number of bedrooms in houses in the same city;
- ***Area Population***: population of the city where the house is located;
- ***Price***: the price at which the house was sold;
- ***Address***: the address of the house.

Exploratory Data Analysis

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```
USAhousing = pd.read_csv('USA_Housing.csv')
```

```
USAhousing.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanielstown, WI 06482...
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5000 entries, 0 to 4999
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	Avg. Area Income	5000 non-null	float64
1	Avg. Area House Age	5000 non-null	float64
2	Avg. Area Number of Rooms	5000 non-null	float64
3	Avg. Area Number of Bedrooms	5000 non-null	float64
4	Area Population	5000 non-null	float64
5	Price	5000 non-null	float64
6	Address	5000 non-null	object

```
dtypes: float64(6), object(1)
```

```
memory usage: 273.6+ KB
```

Exploratory Data Analysis

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```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 5000 entries, 0 to 4999  
Data columns (total 7 columns):  
#   Column                                Non-Null Count  Dtype  
---  ---                                -  
0   Avg. Area Income                      5000 non-null   float64  
1   Avg. Area House Age                   5000 non-null   float64  
2   Avg. Area Number of Rooms             5000 non-null   float64  
3   Avg. Area Number of Bedrooms          5000 non-null   float64  
4   Area Population                       5000 non-null   float64  
5   Price                                 5000 non-null   float64  
6   Address                               5000 non-null   object  
dtypes: float64(6), object(1)  
memory usage: 273.6+ KB
```

```
USAhousing.columns
```

```
Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',  
      'Avg. Area Number of Bedrooms', 'Area Population', 'Price', 'Address'],  
      dtype='object')
```

Exploratory Data Analysis

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```
USAhousing.describe()
```

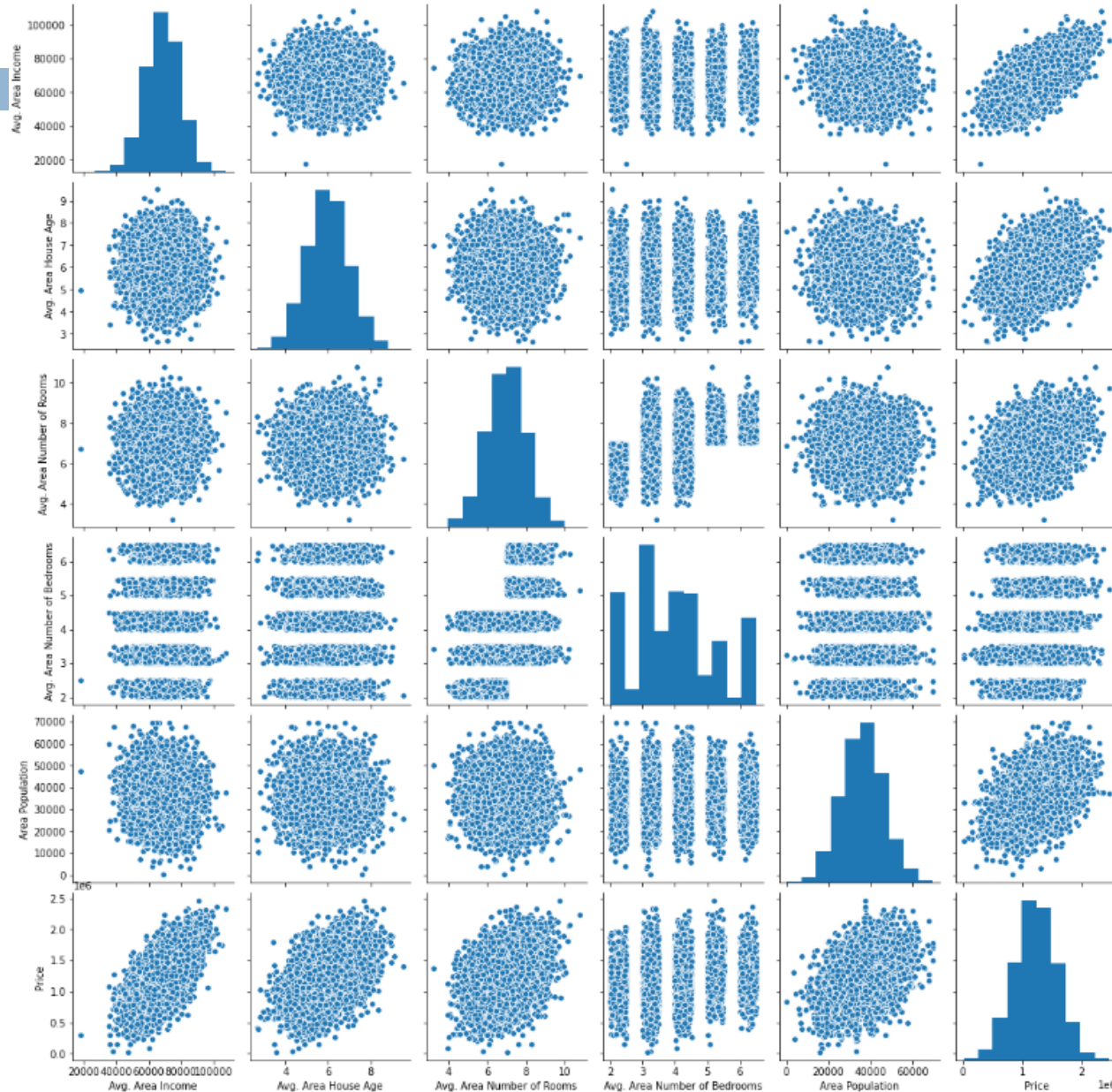
	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	9.975771e+05
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1.232669e+06
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1.471210e+06
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2.469066e+06

Exploratory Data Analysis

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Plot the data to check it.

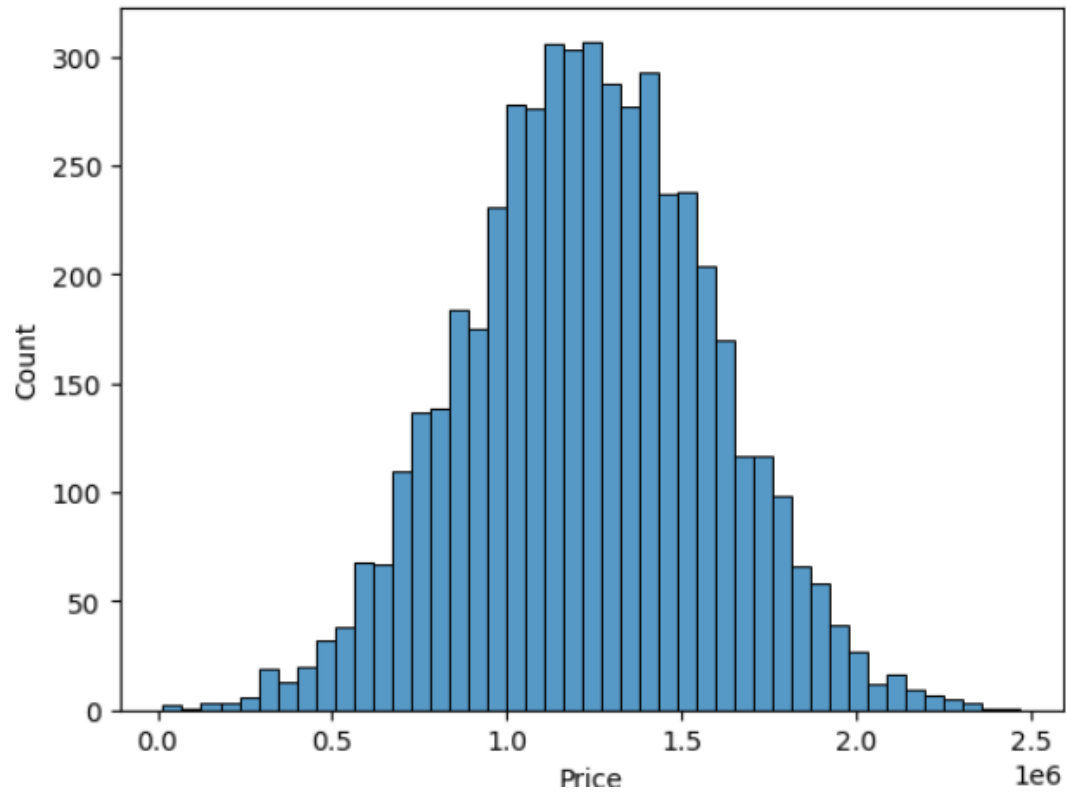
```
sns.pairplot(USAhousing, hue="Price")
```



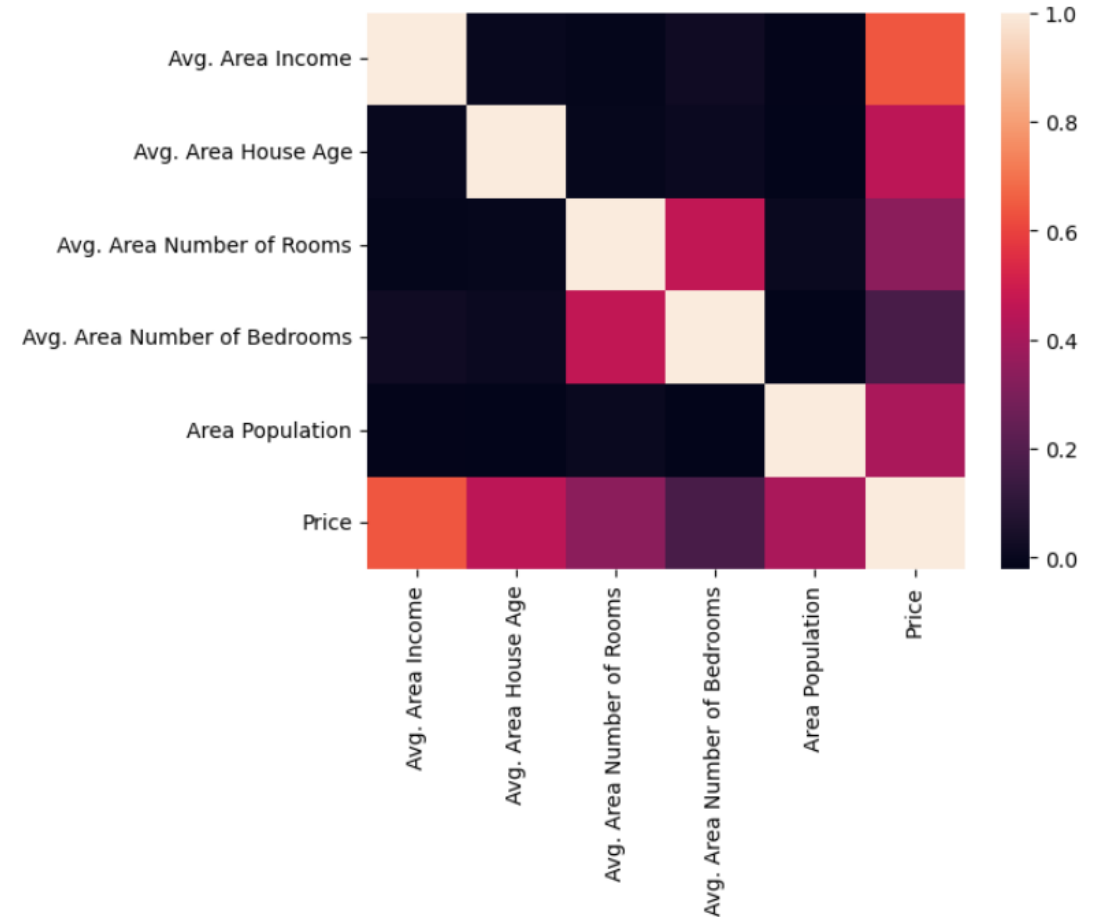
Exploratory Data Analysis

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```
sns.histplot(USAhousing['Price'])
```



```
sns.heatmap(USAhousing.corr(numeric_only=True))
```



Linear Regression

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The target is the **Price**, so we will implement a Linear Regression model.

The feature *Address* will not be considered as it is neither relevant nor a numeric variable.

```
X = USAhousing[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',  
               'Avg. Area Number of Bedrooms', 'Area Population']]  
y = USAhousing['Price']
```

```
from sklearn.model_selection import train_test_split
```

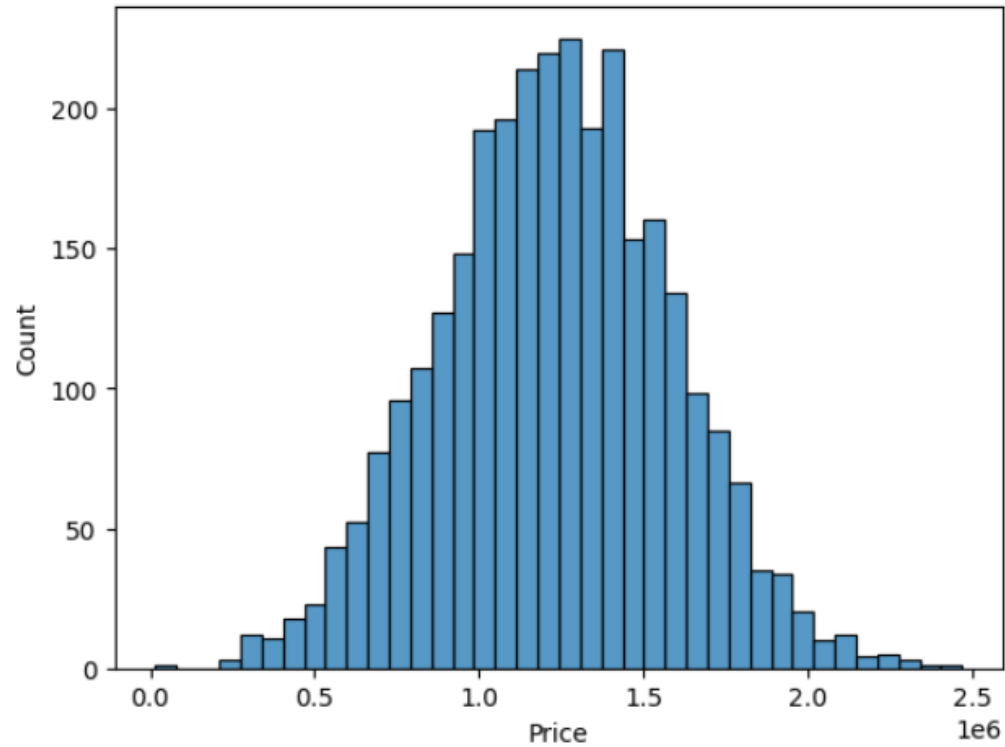
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=101)
```

Linear Regression

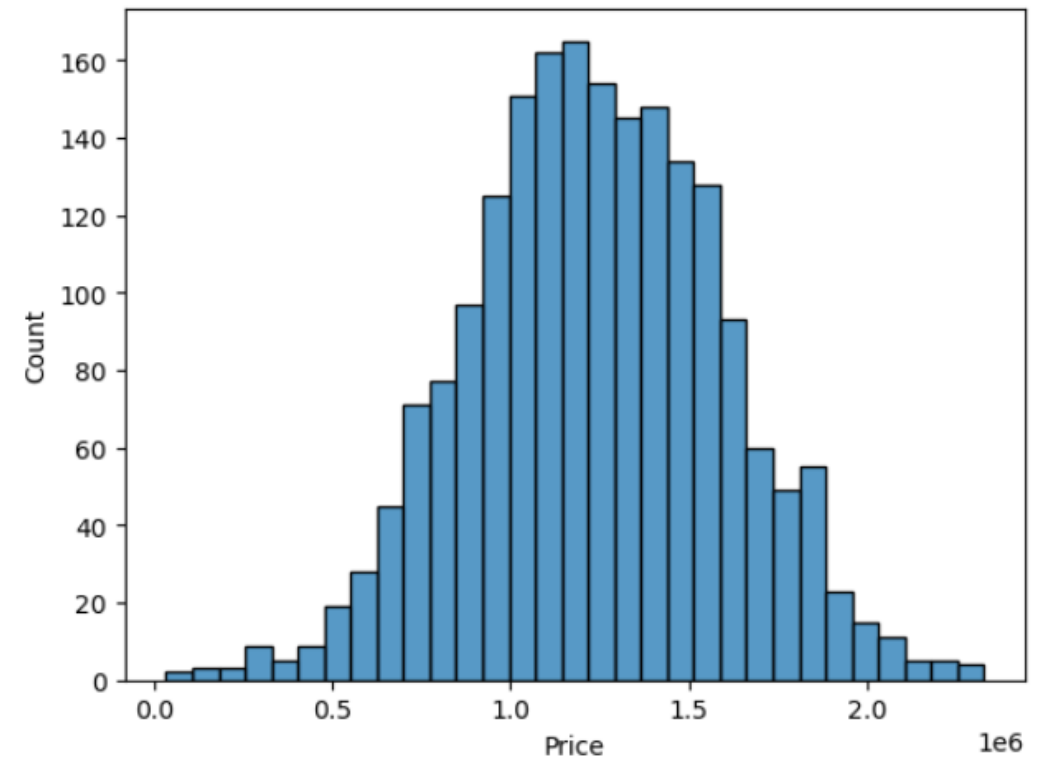
12

Check the distribution of each subset:

```
sns.histplot(y_train)
```



```
sns.histplot(y_test)
```



Linear Regression

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```
sklearn.linear_model.LinearRegression(*, fit_intercept=True, copy_X=True, n_jobs=None, positive=False)
```

Create and train the model:

```
from sklearn.linear_model import LinearRegression
```

```
lm = LinearRegression()  
lm.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?
LinearRegression()

Evaluate the model by checking and interpreting its coefficients:

```
print(lm.intercept_)
```

```
-2640159.7968525267
```

Linear Regression

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```
coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

	Coefficient
Avg. Area Income	21.528276
Avg. Area House Age	164883.282027
Avg. Area Number of Rooms	122368.678027
Avg. Area Number of Bedrooms	2233.801864
Area Population	15.150420

Holding all the other features fixed:

- An increase of 1 unit in **Avg. Area Income** is associated with an increase of \$21.53
- An increase of 1 unit in **Avg. Area House Age** is associated with an increase of \$164883.28
- An increase of 1 unit in **Avg. Area Number of Rooms** is associated with an increase of \$122368.68
- An increase of 1 unit in **Avg. Area Number of Bedrooms** is associated with an increase of \$2233.80
- An increase of 1 unit in **Area Population** is associated with an increase of \$15.15

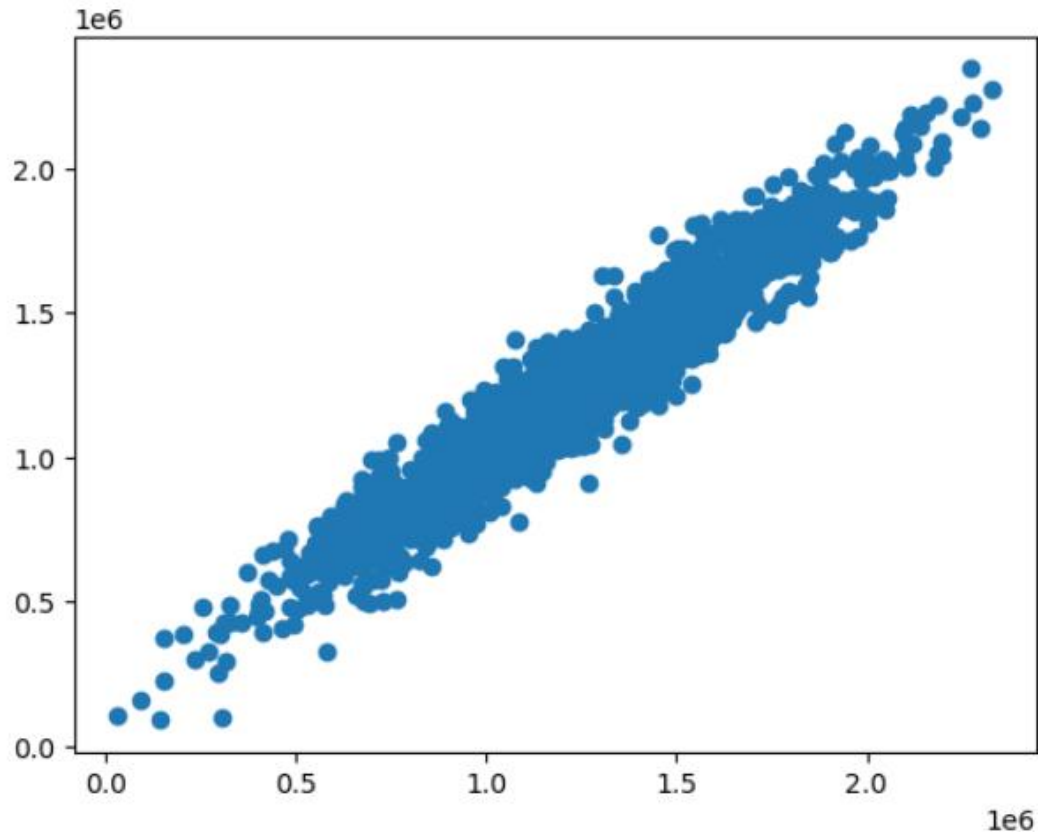
Linear Regression

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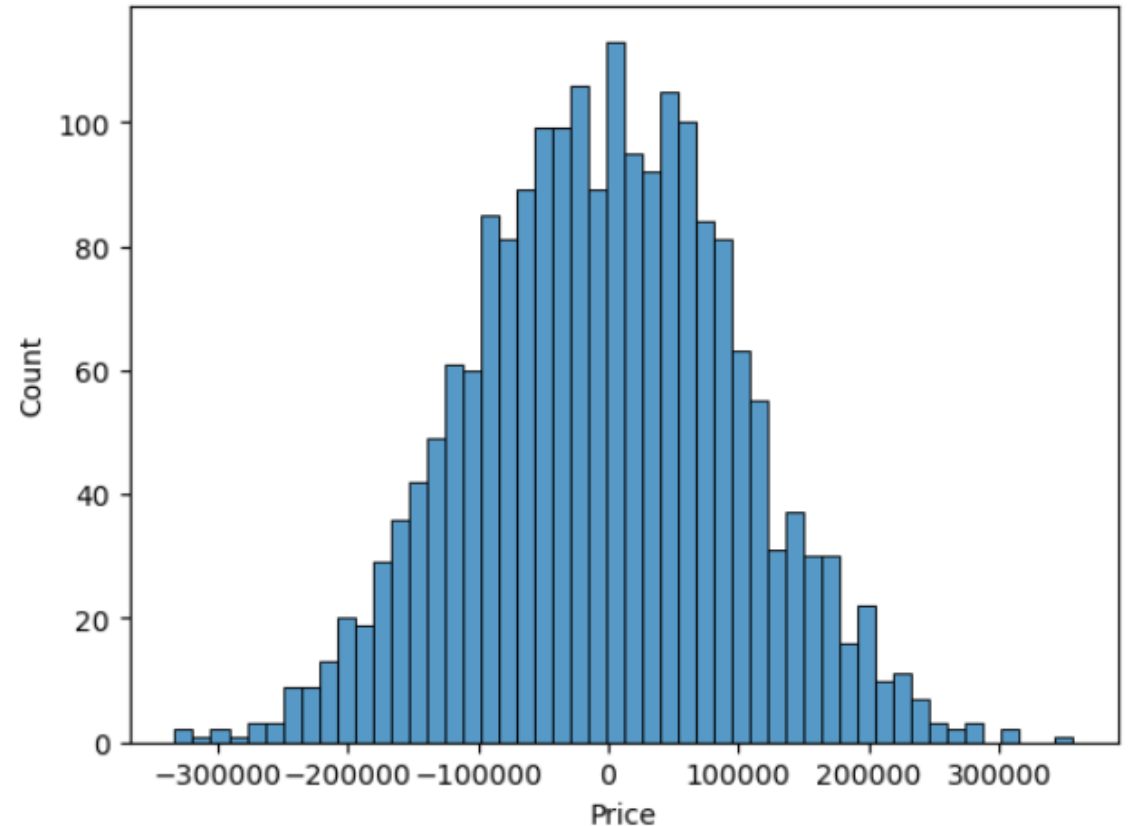
Analyse the predictions and plot it:

```
predictions = lm.predict(X_test)
```

```
plt.scatter(y_test, predictions)
```



```
sns.histplot((y_test-predictions),bins=50);
```



Linear Regression

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Regression Evaluation Metrics

The three most common metrics used to evaluate regression problems are:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Comparing them:

- **MAE** is the easiest to understand because it is the average error;
- **MSE** is more popular than MAE because it “punishes” large errors;
- **RMSE** is even more popular than MSE because it is interpretable in units of the target variable.

All of these are **loss functions** because the aim is to minimise the error.

```
from sklearn import metrics
```

```
print('MAE:', metrics.mean_absolute_error(y_test, predictions))  
print('MSE:', metrics.mean_squared_error(y_test, predictions))  
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
MAE: 82288.22251914947
```

```
MSE: 10460958907.209057
```

```
RMSE: 102278.82922290935
```


Comparing Linear Regression with LASSO

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Regularization

Technique of adding a penalty term to the objective function during training to prevent overfitting and improve generalization performance. The primary goal of regularization is to discourage the model from fitting the training data too closely, which can lead to poor performance on unseen data.

Generate random data and add it to new columns:

```
USAhousing['randomCol1'] = np.random.randint(1, 10, USAhousing.shape[0])
USAhousing['randomCol2'] = np.random.randint(10, 100, USAhousing.shape[0])
```

```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Avg. Area Income                      5000 non-null  float64
1   Avg. Area House Age                   5000 non-null  float64
2   Avg. Area Number of Rooms             5000 non-null  float64
3   Avg. Area Number of Bedrooms          5000 non-null  float64
4   Area Population                       5000 non-null  float64
5   Price                                 5000 non-null  float64
6   Address                               5000 non-null  object
7   randomCol1                            5000 non-null  int32
8   randomCol2                            5000 non-null  int32
dtypes: float64(6), int32(2), object(1)
memory usage: 312.6+ KB
```

Comparing Linear Regression with LASSO

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```
USAhousing.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address	randomCol1	randomCol2
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...	7	46
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...	4	35
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanielstown, WI 06482...	8	49
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820	5	89
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386	7	47

Comparing Linear Regression with LASSO

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```
X = USAhousing[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',  
               'Avg. Area Number of Bedrooms', 'Area Population', 'randomCol1', 'randomCol2']]  
y = USAhousing['Price']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=101)
```

Adjusting Linear Regression

```
lin_reg = LinearRegression()  
lin_reg.fit(X, y)
```

LinearRegression ⓘ ?		
▼ Parameters		
fit_intercept		True
copy_X		True
tol		1e-06
n_jobs		None
positive		False

Comparing Linear Regression with LASSO

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```
sklearn.linear_model.Lasso(alpha=1.0, *, fit_intercept=True, precompute=False, copy_X=True, max_iter=1000, tol=0.0001,
warm_start=False, positive=False, random_state=None, selection='cyclic')

from sklearn.linear_model import Lasso
```

Adjusting LASSO with regularization

```
lasso = Lasso(alpha=2000, random_state=2025)
lasso.fit(X, y)
```

Lasso		
Parameters		
	alpha	2000
	fit_intercept	True
	precompute	False
	copy_X	True
	max_iter	1000
	tol	0.0001
	warm_start	False
	positive	False
	random_state	2025
	selection	'cyclic'

Comparing Linear Regression with LASSO

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Comparing coefficients

```
results = pd.DataFrame({
    'Feature': [X.columns[i] for i in range(X.shape[1])],
    'Linear Regression': np.round(lin_reg.coef_, 2),
    'Lasso': np.round(lasso.coef_, 2)})
```

results

	Feature	Linear Regression	Lasso
0	Avg. Area Income	21.58	21.58
1	Avg. Area House Age	165790.06	163742.88
2	Avg. Area Number of Rooms	120768.29	119163.54
3	Avg. Area Number of Bedrooms	1652.12	949.99
4	Area Population	15.20	15.19
5	randomCol1	220.94	0.00
6	randomCol2	134.50	125.11

Using LASSO, we managed to **eliminate one coefficient** (*randomCol1*).

However, since we have very large coefficients and a small dataset, LASSO did not regularise well.



Logistic Regression

The Problem and the Data

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Problem: development of a Machine Learning model that predicts **which passengers survived** the Titanic shipwreck.

Approach: **Logistic Regression** approach.

Dataset: information about the passengers, including:

- **survival**: whether the passenger survived (0: No, 1: Yes);
- **pclass**: ticket class (1: 1st, 2: 2nd, 3: 3rd);
- **sex**: M: Male, F: Female;
- **Age**: age in years;
- **sibsp**: number of siblings per spouse aboard;
- **parch**: number of parents per children aboard;
- **ticket**: ticket number;
- **fare**: passenger fare;
- **cabin**: cabin number;
- **embarked**: port of embarkation (C: Cherbourg, Q: Queenstown, S: Southampton).

Exploratory Data Analysis

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```
train = pd.read_csv('titanic_train.csv')
```

```
train.head()
```

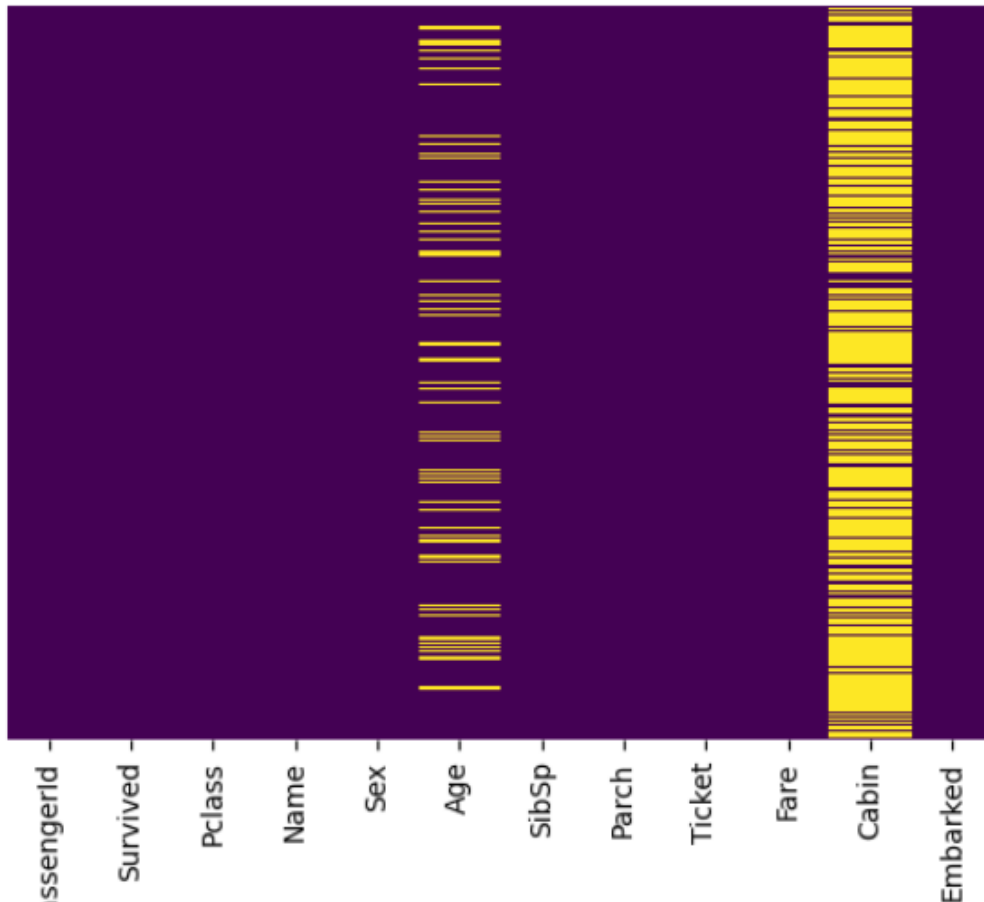
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Exploratory Data Analysis

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Check for missing data:

```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```



Approximately **20%** of the *Age* data is **missing**. This proportion is likely small enough to be reasonably replaced with some form of **imputation**.

However, looking at the *Cabin* column, it appears that we are **missing too much** of that data to do anything useful with it at a basic level. We will probably remove this feature later or replace it with another feature, such as “Cabin Known: 1 or 0”.

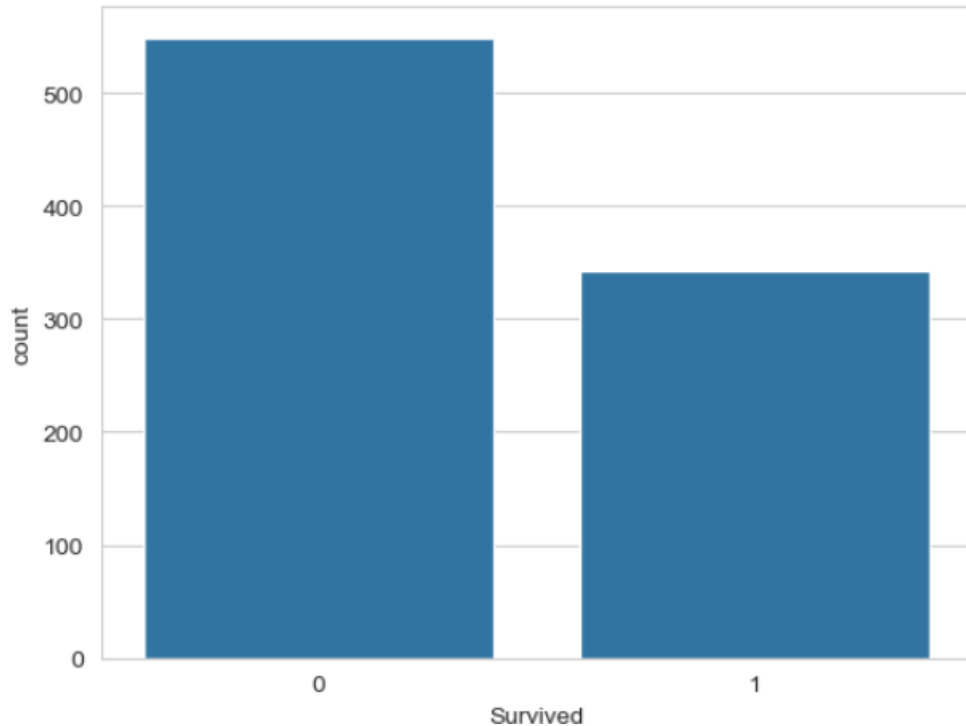
Exploratory Data Analysis

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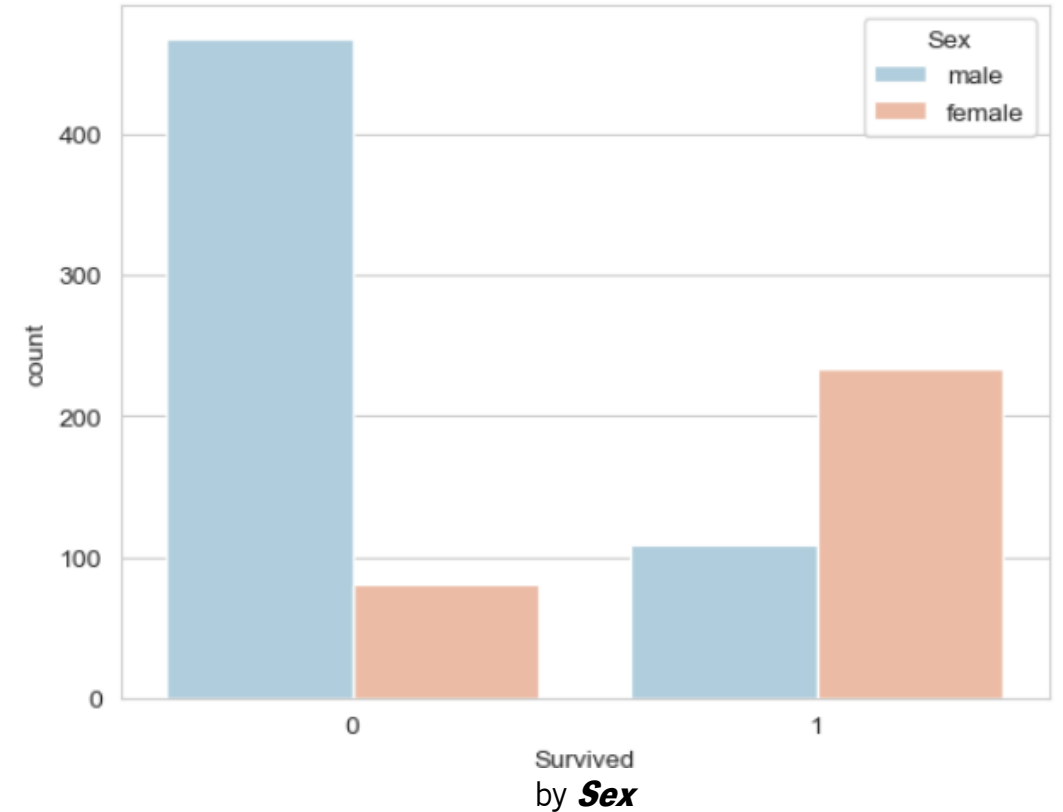
Continue by visualising more of the data.

Survived

```
sns.set_style('whitegrid')
sns.countplot(x='Survived', data=train)
```



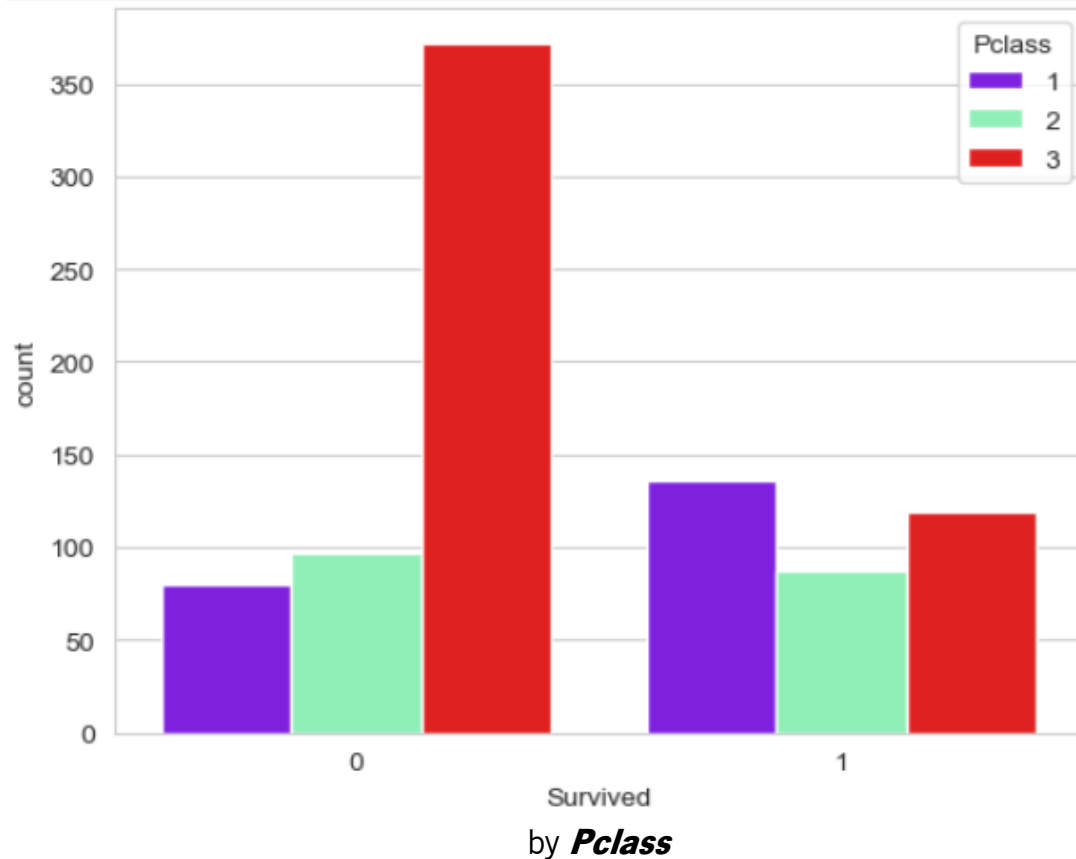
```
sns.set_style('whitegrid')
sns.countplot(x='Survived', hue='Sex', data=train, palette='RdBu_r')
```



Exploratory Data Analysis

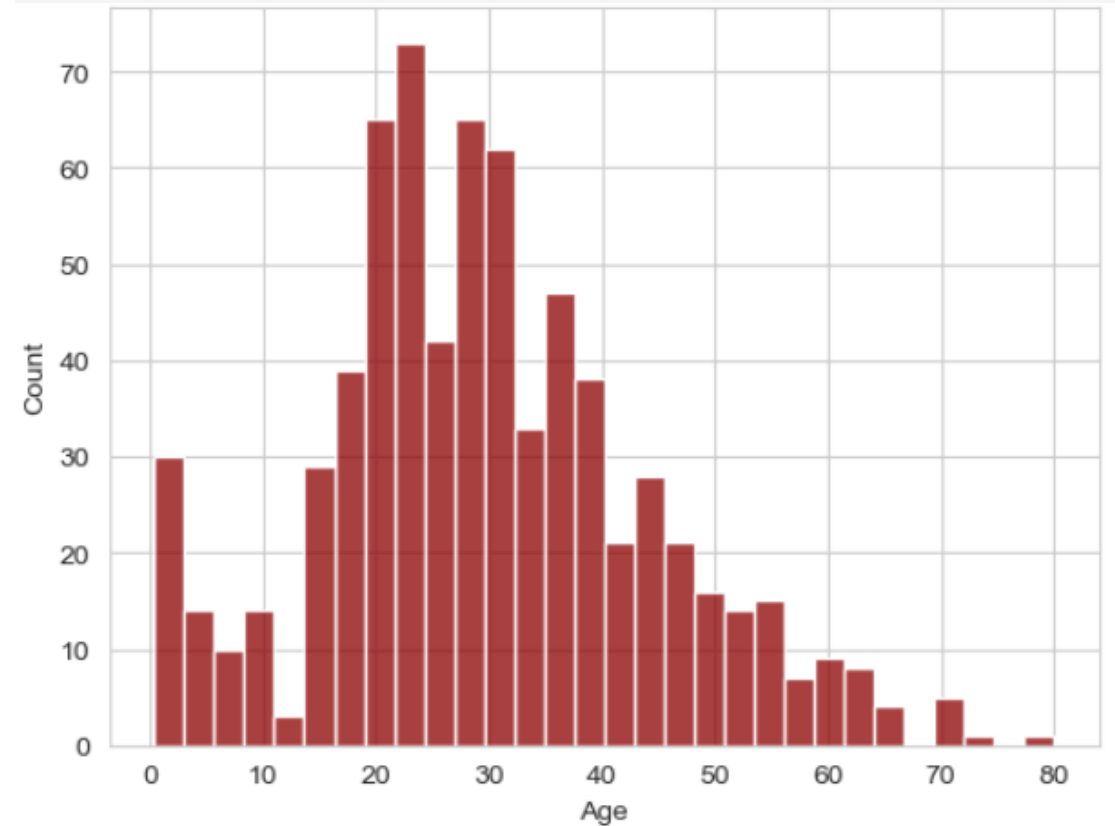
27

```
sns.set_style('whitegrid')
sns.countplot(x='Survived', hue='Pclass', data=train, palette='rainbow')
```



Age

```
sns.histplot(train['Age'].dropna(), kde=False, color='darkred', bins=30)
```

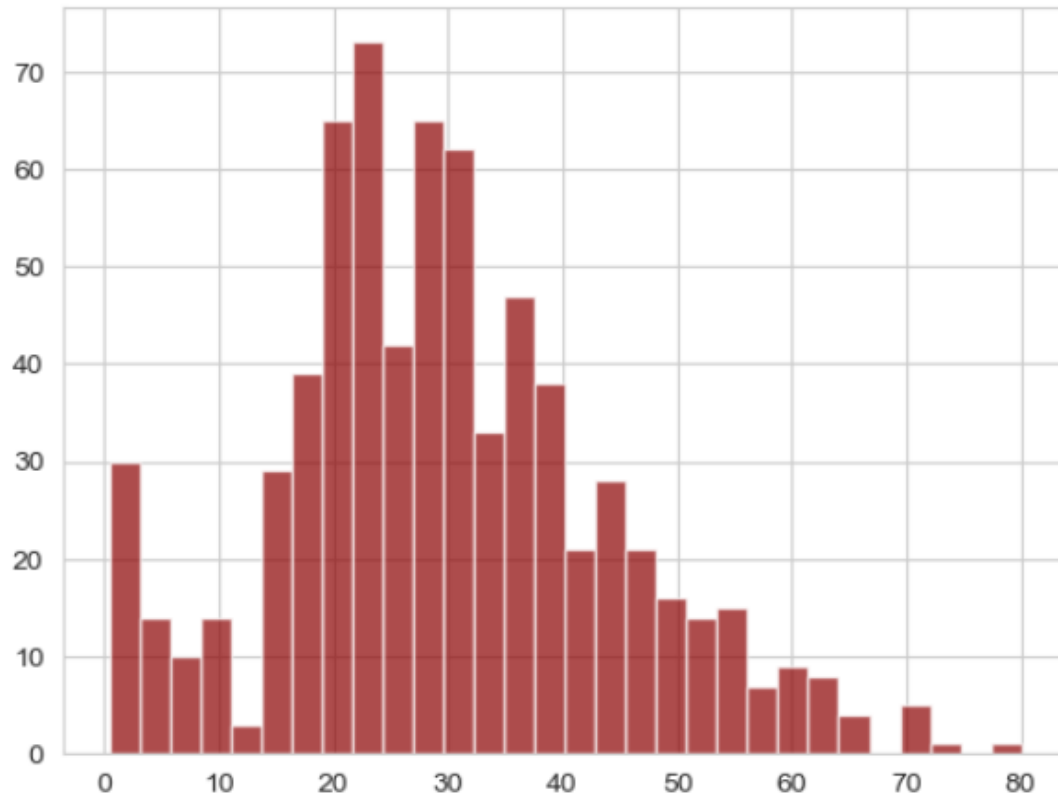


Exploratory Data Analysis

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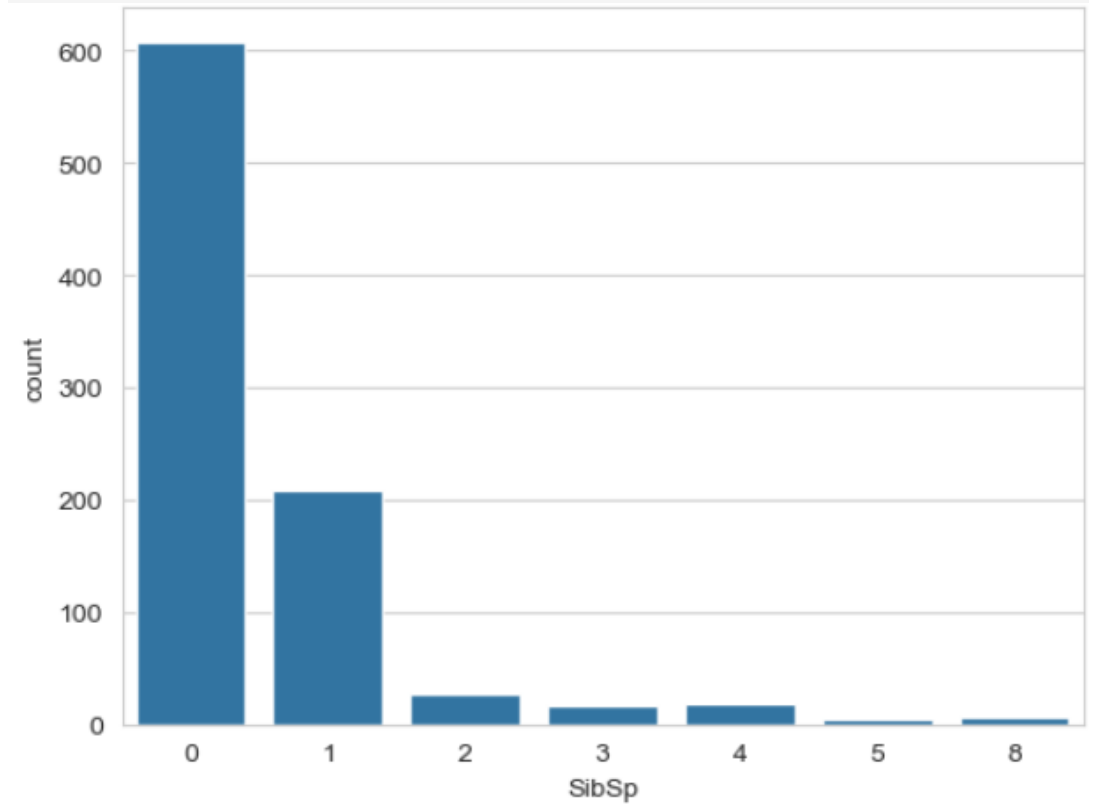
Age

```
train['Age'].hist(bins=30,color='darkred',alpha=0.7)
```



SibSp

```
sns.countplot(x='SibSp',data=train)
```

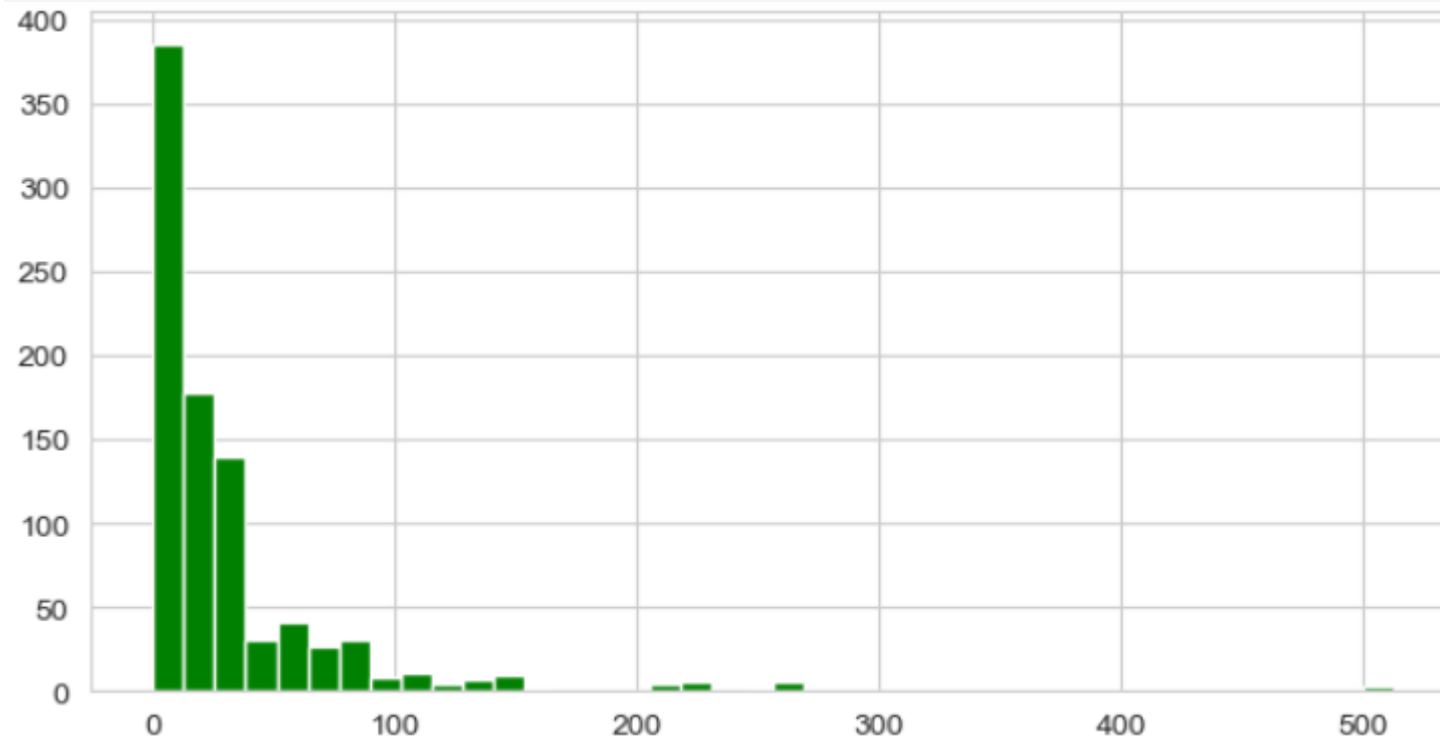


Exploratory Data Analysis

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Fare

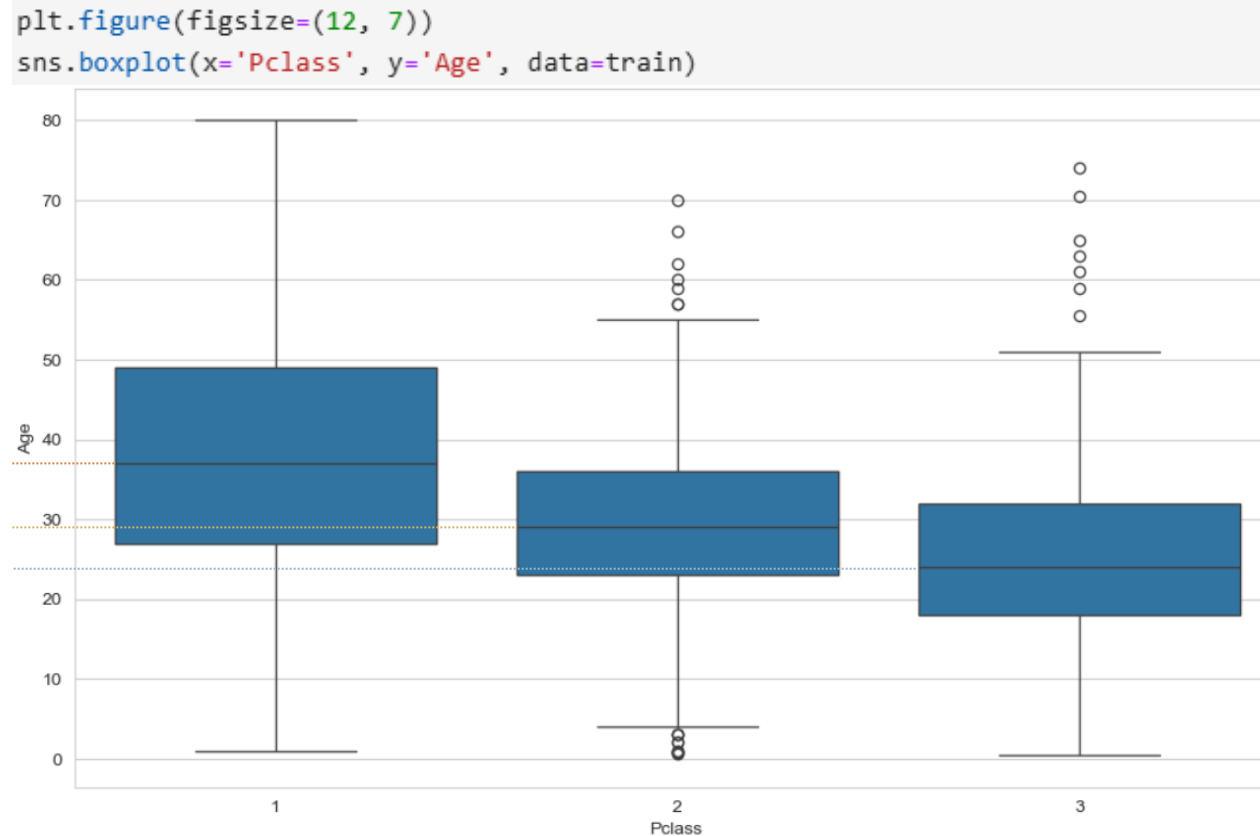
```
train['Fare'].hist(color='green',bins=40,figsize=(8,4))
```



Data Preprocessing

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Instead of removing it, we want to **fill in missing values**. One way to do this is to impute the average age of all passengers (**imputation**).



As expected, we can see that wealthier passengers in the higher classes tend to be older. We will use the **median age values** to impute based on *Pclass*.

Data Preprocessing

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```
def impute_age(cols):
    Age = cols[0]
    Pclass = cols[1]

    if pd.isnull(Age):

        if Pclass == 1:
            return 37

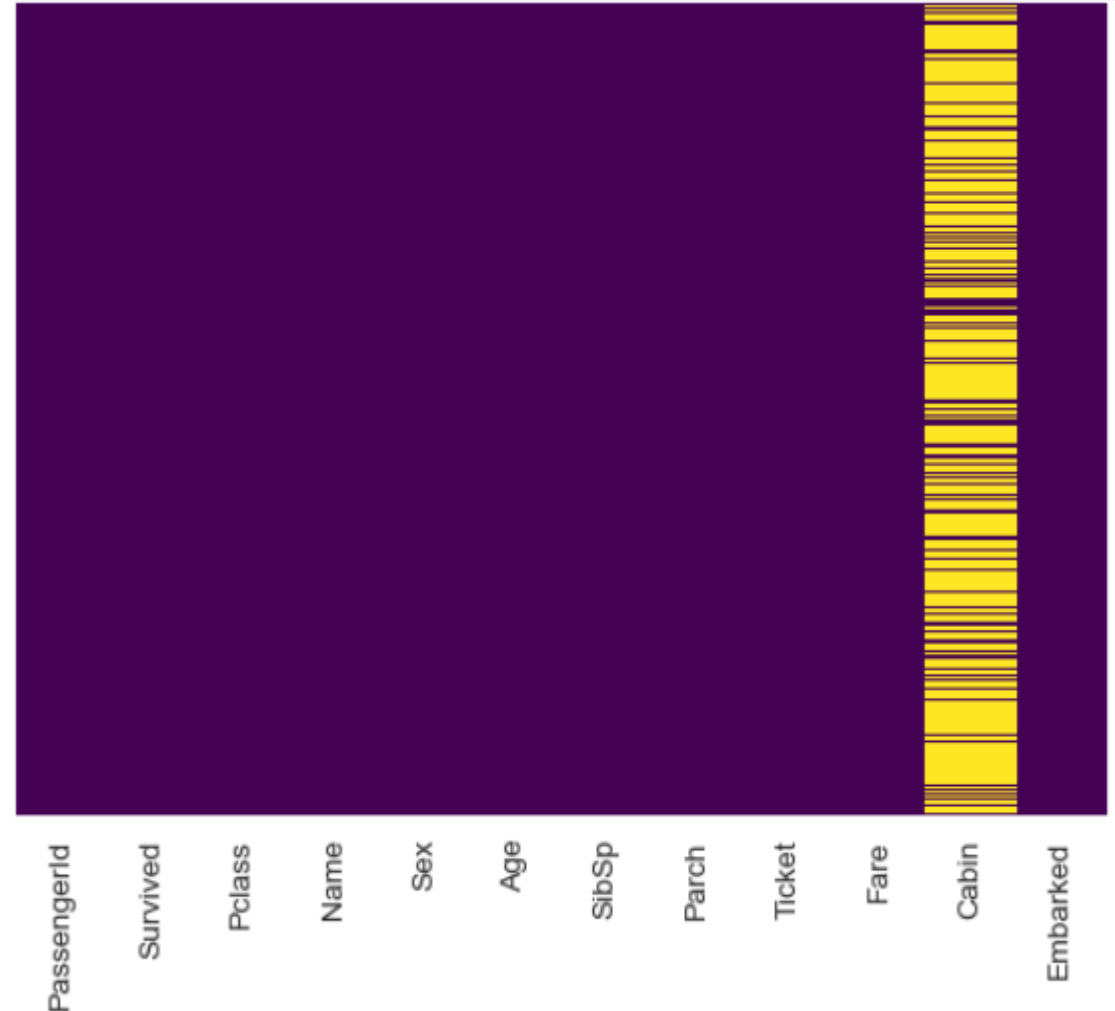
        elif Pclass == 2:
            return 29

        else:
            return 24

    else:
        return Age

train['Age'] = train[['Age', 'Pclass']].apply(impute_age,axis=1)
```

```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```



Data Preprocessing

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Remove the *Cabin* column and the *Embarked* rows that are *NaN*:

```
train.drop('Cabin',axis=1,inplace=True)
```

```
train.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

```
train.dropna(inplace=True)
```


Data Preprocessing

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Use the Pandas library to convert categorical features into dummy variables:

```
train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 889 entries, 0 to 890
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  889 non-null    int64
1   Survived     889 non-null    int64
2   Pclass       889 non-null    int64
3   Name         889 non-null    object
4   Sex          889 non-null    object
5   Age         889 non-null    float64
6   SibSp        889 non-null    int64
7   Parch       889 non-null    int64
8   Ticket       889 non-null    object
9   Fare        889 non-null    float64
10  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(4)
memory usage: 83.3+ KB
```

Sex, Embarked

```
sex = pd.get_dummies(train['Sex'],drop_first=True)
embark = pd.get_dummies(train['Embarked'],drop_first=True)
```

```
train.drop(['Sex', 'Embarked', 'Name', 'Ticket'],axis=1,inplace=True)
```

```
train = pd.concat([train,sex,embark],axis=1)
```

```
train.head()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	male	Q	S
0	1	0	3	22.0	1	0	7.2500	True	False	True
1	2	1	1	38.0	1	0	71.2833	False	False	False
2	3	1	3	26.0	0	0	7.9250	False	False	True
3	4	1	1	35.0	1	0	53.1000	False	False	True
4	5	0	3	35.0	0	0	8.0500	True	False	True

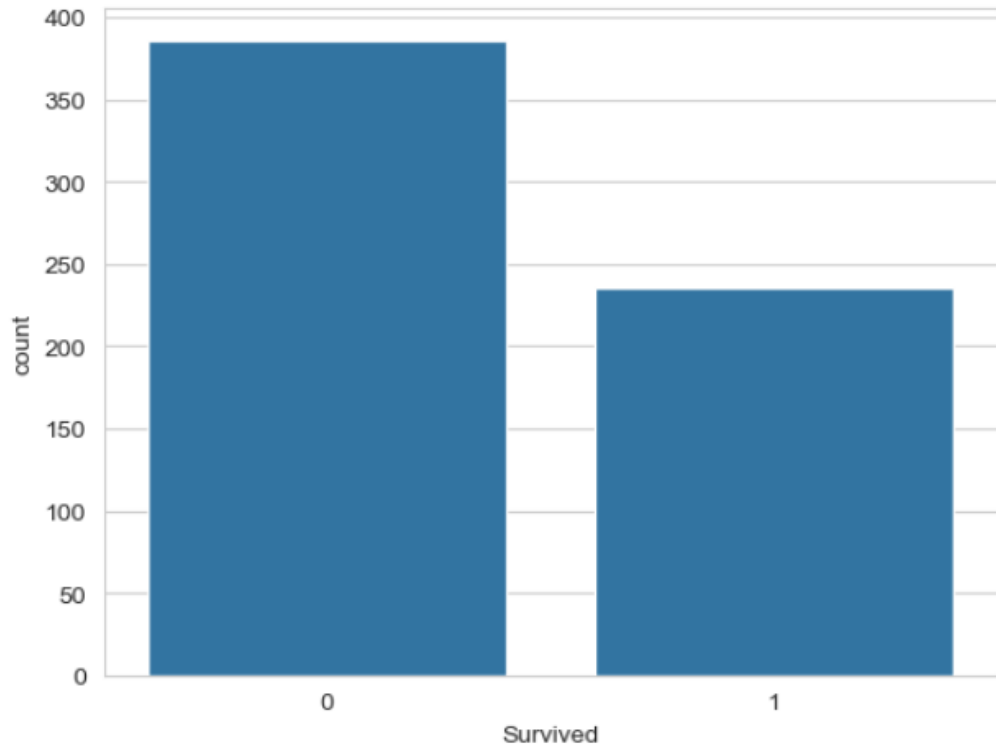
Logistic Regression

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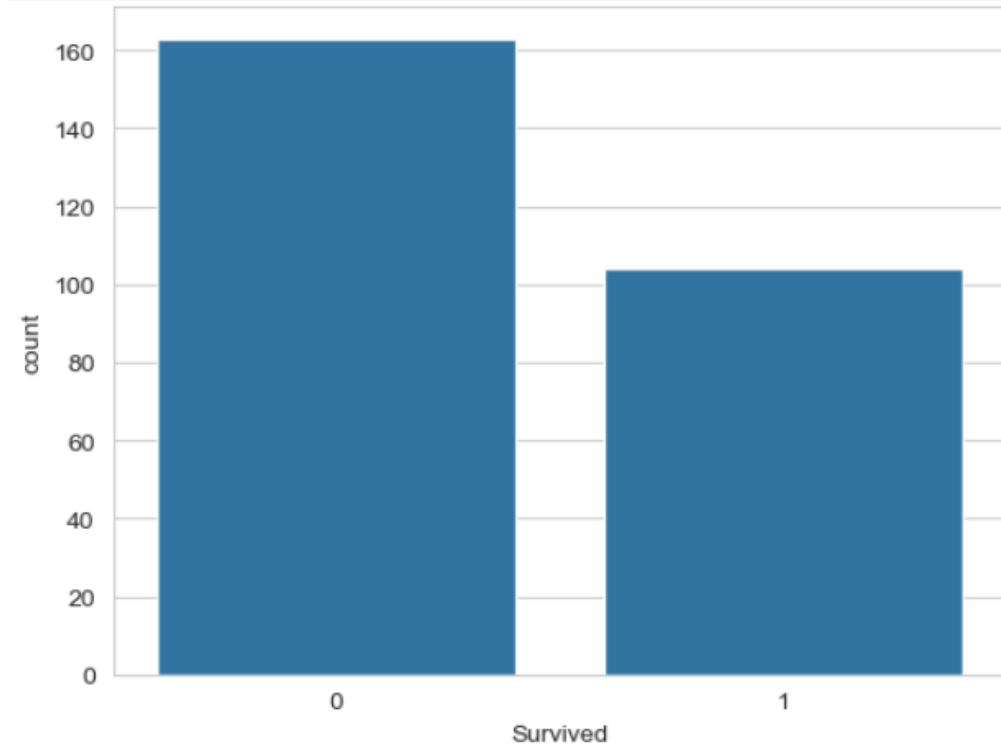
```
from sklearn.model_selection import train_test_split  
X = train.drop('Survived',axis=1)  
y = train['Survived']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, random_state=101)
```

```
sns.set_style('whitegrid')  
sns.countplot(x='Survived', data = pd.DataFrame(y_train,columns=['Survived']))
```



```
sns.set_style('whitegrid')  
sns.countplot(x='Survived', data = pd.DataFrame(y_test,columns=['Survived']))
```



Logistic Regression

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```
sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False, tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None, random_state=None, solver='lbfgs', max_iter=100, multi_class='auto', verbose=0, warm_start=False, n_jobs=None, l1_ratio=None)
```

Logistic Regression **solvers**:

- For small datasets, *liblinear* is a good choice, whereas *sag* and *saga* are faster for larger ones;
- Only *newton-cg*, *sag*, *saga* and *lbfgs* handle multinomial loss for multiclass problems;
- *liblinear* is limited to one-versus-rest schemes.

Supported **penalties** by solver:

- ***newton-cg*** – [*l2*, *none*]
- ***lbfgs*** – [*l2*, *none*]
- ***liblinear*** – [*l1*, *l2*]
- ***sag*** – [*l2*, *none*]
- ***saga*** – [*elasticnet*, *l1*, *l2*, *none*]

```
from sklearn.linear_model import LogisticRegression
```

Logistic Regression

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```
Model 1: random_state = 2022, solver = 'newton-cg'
starttime = time.process_time()

logmodel1 = LogisticRegression(random_state=2022, solver='newton-cg')
print(logmodel1)
logmodel1.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")

LogisticRegression(random_state=2022, solver='newton-cg')
Time spent: 0.015625 seconds

predictions1 = logmodel1.predict(X_test)
```

```
Model 2: random_state = 2022, solver = 'lbfgs'
starttime = time.process_time()

logmodel2 = LogisticRegression(random_state=2022, solver='lbfgs', max_iter=800)
print(logmodel2)
logmodel2.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")

LogisticRegression(max_iter=800, random_state=2022)
Time spent: 0.109375 seconds

predictions2 = logmodel2.predict(X_test)
```

```
Model 3: random_state = 2022, solver = 'liblinear'
starttime = time.process_time()

logmodel3 = LogisticRegression(random_state=2022, solver='liblinear')
print(logmodel3)
logmodel3.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")

LogisticRegression(random_state=2022, solver='liblinear')
Time spent: 0.0 seconds

predictions3 = logmodel3.predict(X_test)
```

Logistic Regression

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Use the following metrics to evaluate the model: *precision*, *recall*, *F1-score* and *confusion matrix*

```
from sklearn.metrics import classification_report, ConfusionMatrixDisplay

print("With 'newton-cg': \n", classification_report(y_test,predictions1))
print("With 'lbfgs': \n", classification_report(y_test,predictions2))
print("With 'liblinear': \n", classification_report(y_test,predictions3))
```

With 'newton-cg':

	precision	recall	f1-score	support
0	0.82	0.91	0.86	163
1	0.84	0.68	0.75	104
accuracy			0.82	267
macro avg	0.83	0.80	0.81	267
weighted avg	0.83	0.82	0.82	267

With 'lbfgs':

	precision	recall	f1-score	support
0	0.82	0.91	0.86	163
1	0.84	0.68	0.75	104
accuracy			0.82	267
macro avg	0.83	0.80	0.81	267
weighted avg	0.83	0.82	0.82	267

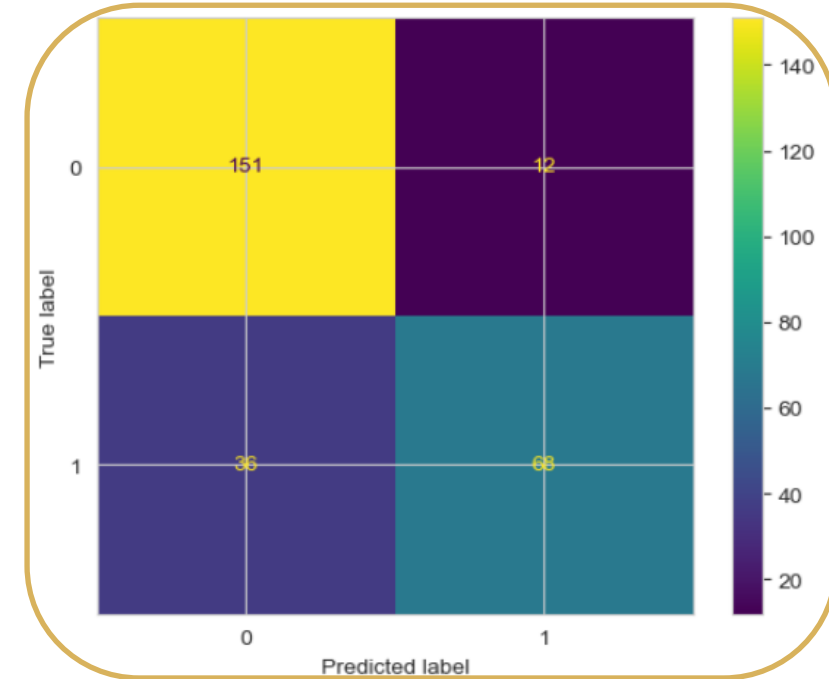
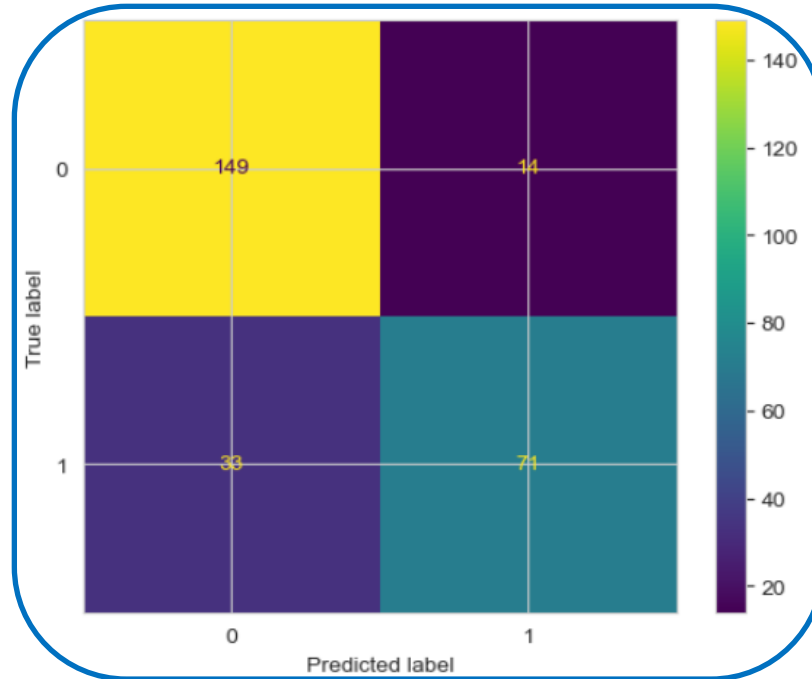
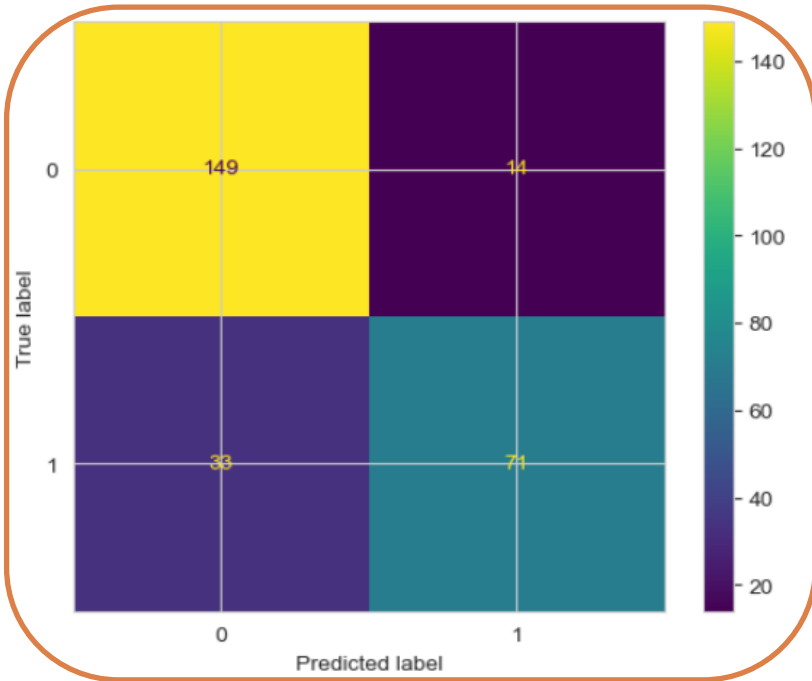
With 'liblinear':

	precision	recall	f1-score	support
0	0.81	0.93	0.86	163
1	0.85	0.65	0.74	104
accuracy			0.82	267
macro avg	0.83	0.79	0.80	267
weighted avg	0.82	0.82	0.81	267

Logistic Regression

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```
ConfusionMatrixDisplay.from_predictions(y_test, predictions1)  
ConfusionMatrixDisplay.from_predictions(y_test, predictions2)  
ConfusionMatrixDisplay.from_predictions(y_test, predictions3)  
plt.show()
```





Hands On